Bayesian Persuasion Mediating Cheap Talk

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Abstract

We study a multi-player model of sequential communication. A receiver obtains information from two sources. The first is cheap talk with a sender, who acquires the information, while the second is Bayesian persuasion with a mediator, who modifies the initial information. We prove that Bayesian persuasion does not defy truthful communication in cheap talk, but reduces the ex-ante utility of the sender. However, the fineness of the state-space partition in cheap talk can be altered to mitigate utility losses. We also demonstrate that a signalling rule in the persuasion is defined by whether the revealed partition element triggers an action of the receiver that is unfavourable to the mediator. In addition, we characterise preferences of the mediator regarding a cheap-talk partition formed by the sender. Finally, we illustrate our results drawing on a communication process between tax authorities.

Keywords: Cheap talk; Bayesian persuasion; mediation; tax avoidance.
JEL classification: C72, D82, D83, H26.

1 Introduction

Actual information exchange often combines different modes of communication. A persuading agent conveys information to another agent convincing him to undertake an action that is beneficial to her. Meanwhile, the information revealed by the persuader may originally stem from a strategically behaving sender and be available to both the persuader and the ultimate receiver. In particular, the initial sender may introduce a strategic noise, because of a difference in preferences that the ultimate receiver and she might have. Hence communication strategy of the persuader must be consistent with the information revealed by the sender. Specifically, the noise introduced on the preceding communication stage affects the efficiency of the following persuasion of the ultimate receiver.

A real-world example for the described situation is the interaction between a taxpayer, domestic and foreign tax authorities. The taxpayer undertakes business activities on the foreign territory and must report her foreign source income to the domestic tax authority. The taxpayer would prefer the authority to accept a reported value of taxable income. However, depending on her propensity for tax avoidance, she would prefer if the tax agency underestimates her income, because excessive tax burden makes her worse off. Meanwhile, the foreign tax authority might benefit from tax-planning activities, because it attracts taxes and enlarges demand for business services. Hence the foreign agency may provide additional information, which is in line with the taxpayer's report, to adjust the expectations the domestic tax authority has on the taxable income. The coincidence of preferences encourages aggressive tax planning.

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Such a coupled communication can be modelled pairing together cheap talk and Bayesian persuasion. However, there is a lack of papers that investigate this combination. To the best of our knowledge, the only available paper so far was produced by Jain (2018). The topic of her work is a recruiting process of fresh graduates, where a school simultaneously provides to prospective employers two pieces of information on its graduates. The first is a binary examination mark, while the other is a reference letter about the graduate's soft skills, which comes as well in the form of a binary variable. Conceptually, the former communication process is modelled as Bayesian persuasion, while the latter is modelled as cheap talk. The results of the paper demonstrate that the availability of a cheap-talk message decreases information value of the examination mark in distinguishing between two graduate types.

Despite some advancement in the field, we need a model of information mediation with separated roles of the cheap-talk sender and the persuader. In addition, good real-world implementation requires the advantage of the cheap-talk models to communicate noisy information. In this paper, we developed a model of mediated communication, where the cheap-talk sender reveals realised state with a strategic noise and the persuader applies a signalling rule to adjust the expectations of the receiver. For the cheap-talk communication we adopted the seminal approach developed by Crawford and Sobel (1982). To keep the analysis tractable we substituted the quadratic by the piecewise linear utilities proposed by Dilme (2018). The distinctive feature of this form of utility is that the agent's preference is defined not by a shift in the state space termed bias, but by the slope of her utility function.

Next, we integrate the Bayesian persuasion from Kamenica and Gentzkow (2011) into the model not just as an additional source of information, but as a modifier of the cheap-talk communication outcomes. The persuader is a strategic mediator, so the communication is defined by the alignment of the agents' interests. However, the continuous state space assumes a restricted set of feasible signalling rules. To deal efficiently with the restriction we converted the geometrical approach developed in Gentzkow and Kamenica (2016) in algebraic form. Despite the general limitations of the method, we could use it to calculate an exact solution due to the assumption of a uniform distribution of states that are common for the cheap-talk models. As a result, our model of mediated communication naturally integrates cheap talk and Bayesian persuasion providing closed solutions suitable for algebraic analysis.

The developed model demonstrates benefits the persuader has from the mediated communication. While she must truthfully reveal realised signals according to the signalling rule of choice, the cheap talk reduces the feasible state space that can improve efficiency of the persuasion. We derive a condition that indicates, when the preceding communication makes the persuader better off. However, the condition holds only, when the realised state triggers unfavorable action of the ultimate receiver. The cheap-talk sender is worse off, but to the extent that keeps the communication truthful. The sender can improve her payoff by change in fineness of the partition of the state space. The more the sender's utility is skewed, the less it can be trusted. Hence, being less trusted, she becomes more vulnerable to Bayesian persuasion. In the motivating example, for instance, tax evaders attract tax harbours.

We should mention some papers to reveal the context of the topic. Blume et al. (2007) research the case of cheap talk, where a partition element is revealed with random error. By contrast, we consider strategic meddling into communication. Both Ivanov (2010) and Ambrus et al. (2013) analyse a mediator, who reveals partition elements applying mixed strategies. In our model the mediator modifies beliefs within a revealed partition element. Hedlund (2017) investigates Bayesian persuasion with a privately informed senders. We focus on the case, where a revealed partition element is public information. In addition, Ichihashi (2019) investigates the effect of an exogenously restricted set of signals available for persuader, while Li and Norman (2018) research sequential persuasion. Finally, Lipnowski and Ravid (2020) show difference in the sender's payoff between Bayesian persuasion and cheap talk, but not their interaction.

The remainder of the paper is organised as follows. The next section outlines the structure of the model. Section 3 demonstrates results on optimal partition in cheap talk and signalling rule in Bayesian persuasion. Section 4 provides a welfare analysis and stipulates the condition that makes Bayesian persuasion beneficial for the persuader. The paper concludes with a summary and directions for further research. All proofs are provided in the Appendix.

2 Setup

The model includes a set of three players $\{s, p, r\}$ with unique roles: sender, persuader, and receiver. There is a state of the nature θ, which is a stochastic variable with the uniform distribution over the state space $\Theta := [0, 1]$. A realised state $\widehat{\theta}$ is private information of the sender, while other agents have beliefs on the state μ_i that is some density $\Delta(\Theta)$ with the mean m_i for $i \in \{p, r\}$. The sender's utility (1) is a piecewise linear function that depends on a mean of the receiver's belief. The utility increases with m_r being close to θ, but the slopes of the utility are different, depending from which side the mean approaches θ. For illustrative purpose we normalise the slope on the domain $(\theta, 1]$ to unity, while the slope on the domain $[0, \theta]$ is $b_s \geq 0$. It is worthwhile to mention, that the utility is motivated by the paper of Dilme (2018).

$$u_s(\theta, m_r) := \begin{cases} -(m_r - \theta) & if\ \theta \leq m_r \\ -b_s(\theta - m_r) & if\ \theta > m_r \end{cases} \tag{1}$$

The persuader's utility (2) is a piecewise linear function similar to the one the sender has, but it includes a shift parameter $a \in \{0, \delta\}$ that is the action of the receiver. Following the motivating example with tax avoidance we are considering a positive $\delta > 0$ that shifts the utility downward for all values of θ, making the persuader worse off. The slope of the utility for states on the left from m_r is normalised to unity. Otherwise, the slope is $b_p \geq 0$ for θ approaching m_r from the right-hand side. The slopes of both the sender and the persuader define their attitude on inconsistency of the receiver's belief. Both prefer the receiver to underestimate the realised state of the nature.

$$u_p(\theta, m_r, a) := \begin{cases} -a - (m_r - \theta) & if\ \theta \leq m_r \\ -a - b_p(\theta - m_r) & if\ \theta > m_r \end{cases} \tag{2}$$

Conversely, the receiver's utility (3) is a step function that depends on the mean of his belief. There exists a threshold level m_r^* for the mean, exceeding which the receiver takes a non-zero action and enjoys a payoff equal to $\delta > 0$. His strategy is linked to the utility, that is $a = \delta$ for $m_r \in (m_r^*, 1]$ and $a = 0$ for $m_r \in [0, m_r^*]$. One can notice, that the payoff structure of the persuader and receiver for given state is a constant-sum game. The receiver's payoff is the exact change of the persuader's payoff. It does not need to hold in general, but it meets the logic of the motivating case. Following the standard assumption of Bayesian persuasion, the tie $m_r = m_r^*$ is broken in favor of the persuader.

$$u_r(m_r) := \begin{cases} \delta & if\ m_r > m_r^* \\ 0 & if\ m_r \leq m_r^* \end{cases} \tag{3}$$

The game includes two communication instances. The first happens between the sender and the receiver, which we modeled as a cheap talk. There exists a partition P of the interval $[0, 1]$ into a set of disjoint sub-intervals $\{p_n\}_{n \in |P|}$ termed *partition elements*. After observing $\widehat{\theta}$, the sender reports to the receiver which partition element the realised state belongs to. Meanwhile, the persuader also receives this information. As usual, the revelation has to be incentive-compatible for the sender. The second communication takes place between the persuader and the receiver, which we model with Bayesian persuasion. In particular, the persuader has to form a signalling rule π as a collection of conditional probabilities $\left\{\Pr[s \mid \theta]\right\}_{s \in S}$ over a set of signals S. Because action a is binary, it suffices to have binary set $S = \{\underline{s}, \bar{s}\}$ that induces two posterior beliefs of the receiver with means $\underline{m}_r \leq m_r^*$ and $\overline{m}_r > m_r^*$.

The standard assumption for Bayesian persuasion is that the persuader truthfully reveals realised signals. In other words, the persuader can deceive, but cannot conceal. Moreover, both the persuader and the receiver are aware of the cheap-talk communication outcomes, so that signals have to be consistent with the revealed partition element. Signals update prior belief $\mu_r(\theta)$ according to Bayes rule forming the receiver's posterior belief as conditional distribution $\mu_r(\theta \mid s)$, that is the nature has state θ given that reported signal is s, while $\mu_r(s)$ is the whole distribution over the states after receiving signal $s \in S$.

$$\mu_r(\theta \mid s) = \frac{\pi(s \mid \theta)\, \mu_r(\theta)}{\int_\Theta \pi(s \mid \theta')\, \mu_r(\theta')} \tag{4}$$

The key restriction for Bayesian persuasion is that posterior beliefs induced by signals have to be Bayes plausible. It implies, that there exists a probability distribution Φ with density ϕ over the set of posterior beliefs $\{\mu_r(s)\}_{s\in S}$, which mean is equal to the receiver's prior belief μ_r.

$$\sum_{s\in S} \mu_r(s)\, \phi\big(\mu_r(s)\big) = \mu_r \tag{5}$$

The timeline of the game encompasses three consecutive stages without repetition. In the first stage the sender chooses partition P, observes $\widehat{\theta}$, and reveals a partition element to the persuader and the receiver. In the next stage, the persuader forms signalling rule π, and reveals a realised signal to the receiver. In the final stage, the receiver forms his posterior belief and chooses action a. Generally speaking, cheap-talk communication happens between the sender and receiver, but the outcomes are reveled publicly. Furthermore the persuader has opportunity to meddle with the receiver's posterior beliefs to improve her own payoff. The solution concept of the game is the perfect Bayesian equilibrium.

3 Results

In the last stage the rule for the receiver's action a is clearly defined by construction, $a = 0$ for $m_r \leq m_r^*$ and $a = \delta$ otherwise. The action depends on the mean of the receiver's belief. Taking into account the standard uniform distribution of the nature, the receiver's prior belief is uniform with the mean at the point 0.5. After the cheap talk, he observes p_n and forms an *interim posterior belief* μ_r^n with the mean m_r^n at the midpoint of the reported partition element. The reason is that any state outside of the reported element is not feasible anymore, because the incentive compatibility constrain for truth-telling (6) guarantees that $\widehat{\theta}$ belongs to the reported interval, while the receiver's interim posterior is still uniform.

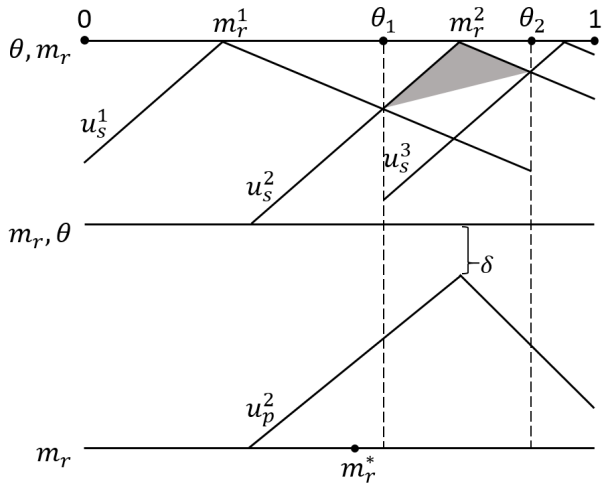

Figure 1: Optimal partition in the cheap talk.

Considering the receiver's mean m_r^n, on the first stage the sender decides on a partition of the state space that must be incentive-compatible with truth-telling. In particular, if $\widehat{\theta}$ is on the boundary of any adjacent elements the sender should be indifferent between reporting either of them, as it is illustrated in Figure 1, so the constraint (6) must hold with equality in this case.

$$u_s\big(\theta, m_r^n\big) \geq u_s\big(\theta, m_r^{n'}\big) \tag{6}$$

$$\theta_{n+1} - \theta_n = b_s\big(\theta_n - \theta_{n-1}\big) \tag{7}$$

The constraint gives the condition (7) that any adjacent element of incentive-compatible partition has to meet. The condition is wholly defined by the slope of the sender's utility.

The skewed nature of the sender's utility leads to difference in lengths of any adjacent intervals of the partition. The smaller the slope b_s, the larger the difference between adjacent elements, consequently, the less partition is informative on small values of the nature. Fully truthful cheap-talk communication is possible only for $b_s = 1$, while $b_s > 1$ does not improve communication, but changes the backward partition depicted in Figure 1 to forward partition. In the latter the finest partition is on values of θ that are closer to 0. This property is general for cheap talk, when

fineness of partition drops in the direction that is preferable by the sender, because she has more incentive to misinform receivers there.

The mean of the receiver's interim posterior belief is the midpoint of the revealed partition element, but it does not always coincide with the persuader's preferences. The reason is that the persuader's utility is also skewed. She does not know the realised state, but in the reported partition element she prefers to maximise exposure of the piece of her utility function with the smaller slope. Let $\widehat{y}_n$ be an optimal value of m_r for a given partition element $p_n := [\theta_{n-1}, \theta_n)$ that benefits the persuader most. It is obvious, that the persuader's optimum tends to get closer to the lower boundary of the partition element. The smaller the slope b_p, the larger the difference between the receiver's interim posterior m_r^n and the *persuader's preference* $\widehat{y}_n$.

$$\widehat{y}_n = \frac{1}{1+b_p}\theta_{n-1} + \frac{b_p}{1+b_p}\theta_n \tag{8}$$

The persuader's preference comes close to the mean of the interim posterior, when the slope b_p approaches 1, unless the reported partition element includes the threshold m^*. In this case the effect of the action a is another reason for manipulation with the receiver's beliefs. Considering the uniform prior, in Bayesian persuasion the receivers accepts a Bayes plausible signal and updates his prior according to Bayes rule (4) forming his posterior belief. After partition element p_n is reported, a rational player expects zero probability for any states that are outside of the interval, that is $\Pr\big[\theta \notin p_n\big] = 0$. Since the distribution of the states is uniform, it increases the density of the prior belief from 1 to $1/|p_n|$. Another consequence is that the persuader cannot induce any signal that forms posterior beliefs outside of a *compactified reported interval*, which is $p_n = \big[\theta_{n-1}, \theta_n\big]$. Compactification is possible due to indifference of the sender on boundaries of partition elements. Henceforth, p_n will define both partition elements and their compactification.

Before we move to Bayesian persuasion, it is worth to consider effects of changes in posterior on optimal partition in preceding cheap talk. According to the incentive-compatibility constraint, payoff on the boundaries between adjacent elements is identical. Because of a single-picked feature of the utility, any convex combination of utilities with different posteriors cannot provide a payoff lower than a convex combination of the payoffs on the boundaries. In Figure 1 possible payoffs for any realisation of the state in p_n are depicted by the grey area. As a result, for any realisation of the state in the reported element the convex combination of utilities within the element provides higher payoff than reporting any other partition elements.

LEMMA 1. For given p_n any Bayes plausible signalling rule that induces posterior beliefs that $m_r(s) \in p_n$ for all $s \in S$ does not violate the incentive-compatible constraint for the sender.

The reasoning behind the lemma can be described in two steps. Consider the case with $\widehat{\theta} = \theta_{n-1}$, which is the boundary between partition elements p_{n-1} and p_n. Partition element p_n is reported and the sender's utility in this interval of the domain is a linear function (1). Considering risk neutrality of the player, it is clear that any convex combination of beliefs inside the interval will not change the sender's utility. The next step is to consider any $\widehat{\theta} \in p_n$, but not on the boundary. In this case, the additive property of the linear function helps. Moving the unique maximum into the interior of the interval reduces the payoff of a low signal, but the utility in the adjacent interval decreases by the same value. Meanwhile, payoff for any high signal increases, so that a convex combination of payoffs in this interval, which is Bayes plausible is large enough. This holds similarly true for another partition element p_{n+1} that is adjacent to p_n.

Since both communication instances are compatible, it is possible to define Bayesian persuasion on the second stage. There can be two possible cases regarding reported p_n, namely either $m_r^* \notin p_n$ or $m_r^* \in p_n$. For the latter the receiver chooses $a = \delta$, if $m_r^n > m_r^*$. It increases her payoff, but reduces the utility of the persuader. However, the persuader can apply a non-trivial signalling rule in both cases to improve her expected payoff. We call a signalling rule *trivial*, when $\underline{m}_r = \overline{m}_r = m_r^n$. The interim posterior mean m_r^n of the receiver is equal to $(\theta_{n-1} + \theta_n)/2$, which can be modified by applying a signalling rule and inducing at least two different posterior means. Below, we consider payoff outcomes of Bayesian persuasion in both mentioned cases illustrated in Figure 2.

We begin with the case when $m_r^* \notin p_n$. The persuader can try to apply a signalling rule and form a convex combination of her two utility functions. In particular, for any state realisation of a given partition element one utility will be used with probability ϕ and another utility function will appear with the complementary probability $1-\phi$. The former could have a maximum at $m_r(s) = \widehat{y}_n$ defined in (8), while the maximum of the latter is at any point on the right of the interim posterior mean. As a result, the *expected utility* function of the persuader is $\nu_p = \mathbb{E}_\phi\big[u_p\big(m_r(s)\big)\big]$. However, such a strategy is not beneficial for the persuader. The reason is the Bayes plausibility (5), when convex combinations of posteriors have to be equal to the interim posterior, which serves as a prior to the Bayesian persuasion. Regarding utilities, this means that for any $\theta \in p_n$ values of the utility functions with maximums at points $\underline{m}_r$ and $\overline{m}_r$ is at most equal to the utility with the maximum at the interim posterior.

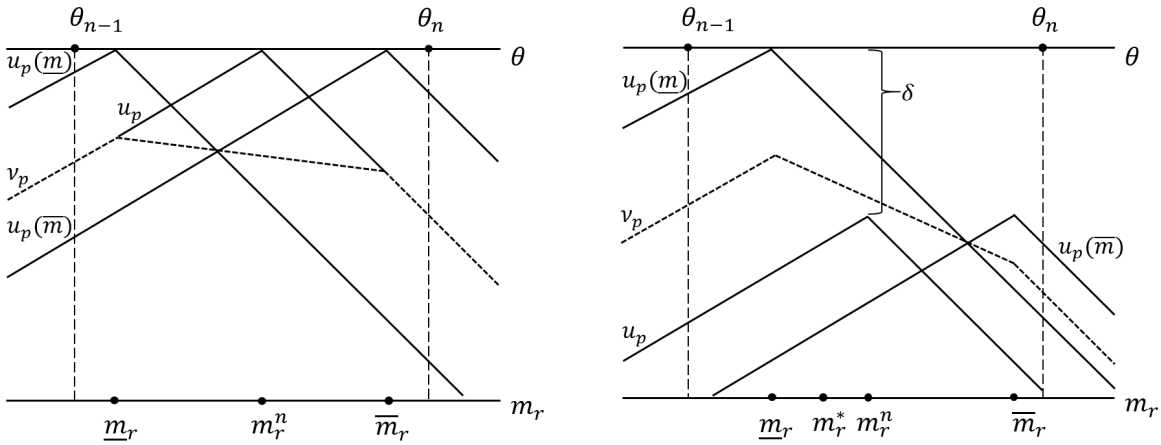

Figure 2: Bayesian persuasion when $m_r^* \notin p_n$ (left) and $m_r^* \in p_n$ (right).

Taking into account that all three utilities are similar piecewise linear functions, there are three intervals to consider. In the intervals $\big(\theta_{n-1}, \underline{m}_r\big]$ and $\big[\overline{m}_r, \theta_n\big]$ the convex combination of utilities $u_p(\underline{m}_r)$ and $u_p(\overline{m}_r)$ induced by signals $\underline{s}$ and $\bar{s}$, respectively, coincide with $u_p(m_r^n)$ due to the Bayes plausibility. However, in the interval $\big(\underline{m}_r, \overline{m}_r\big)$ the expected utility is strictly below of any of two persuader's utilities. The reason is that utility $u_p(\underline{m}_r)$ decreases within the interval, while utility $u_p(\overline{m}_r)$ increases. As a convex combination of these utilities, the expected utility must fall below $u_p(m_r^n)$. Hence, for any value θ in partition element p_n the values of ν_p are at most the values of utility $u_p(m_r^n)$. The persuader will always apply the trivial signalling rule, if $m^* \notin p_n$, but in another case she could benefit from Bayesian persuasion.

For the same reason Bayesian persuasion does not work, if $m^* \in p_n$, but $m_r^n \leq m^*$. The important case to consider is $m_r^n > m^*$. The receiver plays $a = \delta$ and the persuader's utility decreases by δ. This shift in payoffs changes the outcome of Bayesian persuasion. It might be the case that values of ν_p are at least equal to values of the utility based on the interim posterior for all states in the reported partition element, as depicted at the rights side of Figure 2. However, it is the ultimate scenario. Since neither the persuader nor the receiver knows the realised state, it is enough to meet the condition (9), when the expected value of ν_p is larger than the expected value of $u_p(m_r^n)$. The expected value is defined with respect to $\theta \in p_n$ that has a uniform distribution $uni(p_n)$.

$$\mathbb{E}\big[\nu_p(\theta) \mid \theta \sim uni(p_n)\big] \geq \mathbb{E}\big[u_p(\theta, m_r^n) \mid \theta \sim uni(p_n)\big] \tag{9}$$

The shift δ and the slope of the utility b_p are the main parameters that define whether Bayesian persuasion is beneficial for the persuader. However, to derive a closed solution we need to define explicitly the expected utility ν_p, what requires posterior beliefs and Bayes plausible probability ϕ over them in explicit form. The main constraint is that not all means of posterior beliefs are attainable in the case of continuous state space. One way to tackle this restriction is the geometrical representation of the problem developed in Gentzkow

and Kamenica (2016), where feasible signalling rules are constricted between two extreme communication cases, fully informative and babbling. Moreover, it is possible not only to employ this approach, but also to convert it into an algebraic solution due to the specific features of our model.

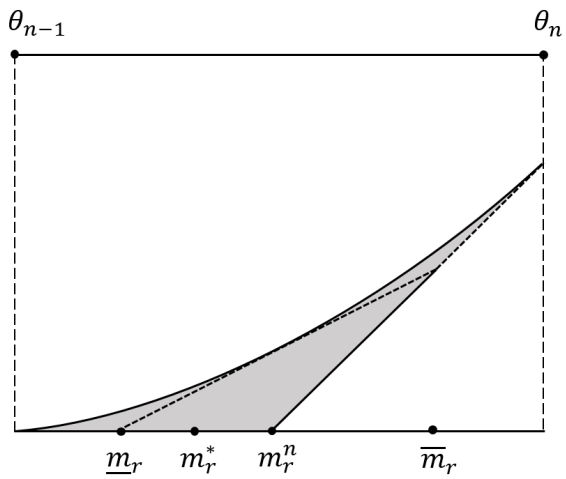

Figure 3: Feasible signalling rules.

A fully informative signalling rule is a uniform distribution, where any posterior mean is revealed with equal probability. Furthermore, babbling is the trivial signalling rule. The second integrals of the two cases are depicted in Figure 3. Note that any feasible signalling rule must be in the shaded area on the picture. Due to $|S| = 2$, the support of the distribution Φ is equal to 2. Hence, the integral of Φ is a piece-wise linear function depicted as the dashed line with the kinks at $\underline{m}_r$ and $\overline{m}_r$.

Since for the trivial signalling rule $\Pr\left[\theta = m_r^n\right] = 1$, its slopes are always 0 on the interval $(\theta_{n-1}, m_r^n]$ and 1 on the interval $(m_r^n, \theta_n]$. Most significantly, the interim posterior mean is always at the midpoint of the p_n. Consequently, for any length of revealed partition element the whole geometric construction will be similar up to the scaling. This consistency of the geometric structure leads to the solution (10), which locks feasible signalling rules to the low posterior mean. The slope on the interval $(\underline{m}_r, m_r^n]$ defines the marginal probability $\phi(\underline{\mu}_r)$, which in our context is beneficial for the persuader, so the slope should be the largest.

LEMMA 2. Given partition element p_n with interim posterior belief m_r^n at the midpoint of p_n and the optimal low posterior mean y^*, the feasible posterior means and the marginal probability ϕ are:

$$\underline{m}_r = y^* \in \left[\theta_{n-1}, m_r^n\right] \; ; \quad \overline{m}_r = y^* + \frac{|p_n|}{2} \; ; \quad \phi(\overline{\mu}_r) = \frac{2(m_r^n - y^*)}{|p_n|} \tag{10}$$

For a given reported partition element the whole signalling rule can be defined by a single variable, that is, the low posterior mean. The geometric approach deterministically binds all elements of signalling rule. The highest marginal probability on low posterior mean can be achieved, when $\underline{m}_r = m_r^*$. However, the skewed property of the persuader's utility (2) makes optima less obvious. In order to define optimal low posterior means y^* we have to solve an optimisation problem with the expected utility $\mathbb{E}\left[\nu_p \mid \theta \sim uni(p_n)\right]$ as object function. However, the solution cannot be lower than the lower boundary of a compactified partition element and can also not be larger then threshold m_r^*. Otherwise, it will trigger the unfavorable action $a = \delta$ that reduces the persuader's utility.

PROPOSITION 1. Given partition element p_n, where the threshold $m_r^* \in p_n$ and $m_r^n > m_r^*$, the persuader chooses signalling rule defined in (10) *and based on the following low posterior mean:*

$$y^* = \begin{cases} \theta_{n-1} & , if \; y \leq \theta_{n-1} \\ m_r^* & , if \; y \geq m_r^* \\ y & , otherwise \end{cases} \qquad , where \;\; y = m_r^n - \frac{|p_n|}{4} - \frac{2\delta}{1 + b_p} \tag{11}$$

The optimum low posterior mean is driven by the shift parameter δ and the persuader's slope. Putting the threshold m_r^* aside, the optimum is bounded inside the first quarter of reported partition element. The less the shift parameter and the stronger the preferences of the persuader, the closer the low posterior to the lower boundary of the partition element. However, one of the consequences of Lemma 2 is that, if $y^* = \theta_{n-1}$,

then $\overline{m}_r = m_r^n$ and $\phi(\overline{\mu}_r) = 1$. This means, that the persuader becomes indifferent between the optimal signalling rule and no Bayesian persuasion at all. Hence we focus on the case with $y^* > \theta_{n-1}$ in further analysis of the welfare of the sender and the persuader.

4 Welfare

After deriving the optimal signalling rule, it is possible to elaborate the condition (9), that makes Bayesian persuasion beneficial for the persuader. The right panel on Figure 2 illustrates a case, when the trivial signalling rule is dominated for any realisation of the state of the nature in the reported partition element. However, the persuader in the second stage is not aware of the realised state and makes a strategic choice regarding her expectation of the state based on its uniform distribution $uni(\Theta)$. Therefore, it is enough to explicitly solve the condition (9) for the signalling rule defined in Proposition 1. Below, we limit the welfare analysis to the case, when the optimal $y*$ is not corner solution at m_r^*.

PROPOSITION 2. Given $m_r^* \in p_n$ and $m_r^n > m_r^*$, the optimal signalling rule defined in (10) *and* (11) *is beneficial for the persuader, if the following condition holds:*

$$\left(\frac{|p_n|}{4} + \frac{2\delta}{1+b_p}\right)^2 \geq |p_n| \frac{2\delta}{1+b_p} \tag{12}$$

The equation (12) demonstrates when the expected utility induced by Bayesian persuasion can provide higher benefits for the persuader. Especially, notice that if $m_r^n \leq m_r^*$ and $\delta = 0$ in the reported interval, that is the receiver chooses action $a = 0$, then the condition does not hold. This reflects our discussion above about the inefficiency of Bayesian persuasion in the case of similar utilities, when the action is not triggered in one of the posteriors. The shift δ has an ambivalent effect on the condition, while its effect is mitigated by the increase in the slope of the utility. The more beneficial the persuader considers lower values of θ, the larger the effect of the shift on the condition.

Another important factor is the relationship between the size of the partition element and efficiency of Bayesian persuasion. The size has a quadratic effect on the condition defined in Proposition 2. The close the partition element, which includes threshold m_r^*, to optimal maximum length $8\delta/(1+b_p)$, the better for the persuader. At the same time, this condition does not fix the place of the partition element in the partition. As far as, the threshold m_r^* belongs to the interval $[y^*, m_r^n)$, persuasion in the element is beneficial. As a result, it is possible to derive a condition that makes it beneficial for the persuader to reduce the interval of possible states from $[0, 1]$ to the smaller $|p_n| < 1$.

COROLLARY. The persuader finds Bayesian persuasion more beneficial after cheap-talk communication, if the following condition holds:

$$1 + b_p < 8\delta \tag{13}$$

According to Lemma 1 on the first stage the sender finds the cheap-talk communication beneficial even if it is followed by the Bayesian persuasion. However, the optimal signalling rule has a negative effect on payoff outcomes for the sender. In fact, if a non-trivial signalling rule is applied, the sender's ex-ante expected utility decreases leaving her worse off. The sender has only one of the cheap-talk variables that can be tuned is the number of partition elements N. Following a typical approach for cheap talk for evaluation of sender's welfare, it is possible to evaluate the ex-ante utility of the sender (14) that is a function of number of partition elements N and the slope of the sender's utility b_s. With increase in the number of partition elements the denominator rises faster than the nominator.

$$\mathbb{E}\big[u_s(\theta, m_r^n) \mid \theta \sim uni(\Theta), n \in \{1, .., N\}\big] = -\frac{(1+b_s)\sum_{n=1}^{N}(b_s^{n-1})^2}{8\big(\sum_{n=1}^{N} b_s^{n-1}\big)^2} \tag{14}$$

The ex-ante utility rises with an increase in number of partition elements. Moreover, condition (7) imposes the restriction only on the size of adjacent partition elements, but not on the size of the first element of partition. Consequently, it is always possible to have a partition with any N by choosing appropriate size of the first element. As a result, it is beneficial for the sender to impose $N = \infty$ for any b_s. In the setup we defined $b_s \geq 0$, but the restriction (7) allows us to generalise the ex-ante utility for both cases $b_s \leq 1$ and $b_s > 1$. The difference between these two cases lies in direction of the partition, either it is forward or it is backward. The former is a reflection of the latter and does not affect ex-ante utility as an integral of the sender's utility over Θ. Hence, we define β as $\beta = b_s$, if $b_s \leq 1$, and $\beta = 1/b_s$ otherwise. Since β is always at most 1, the ex-ante utility converges to a finite number.

$$\mathbb{E}\big[u_s(\theta, m_r^n) \mid \theta \sim uni(\Theta), n \in \{1, .., \infty\}\big] = -\frac{1-\beta}{8} \tag{15}$$

The less the sender is willing to manipulate the receivers beliefs, the closer is the slope b_s to 1 and the higher is the ex-ante utility. In addition, with $N = \infty$ the first partition element (the last for the case with the forward partition) cannot be less than $1 - b_s$. However, the Bayesian communication reduces the ex-ante utility by cutting the expected payoff in the partition element that includes threshold m_r^* and interim posterior encourages non-trivial signalling rule. The sender can decrease the number of elements in partition to make the Bayesian persuasion not beneficial in any partition elements, but it will reduces her ex-ante utility. As a result, the sender faces a trade-off between $N = \infty$, but loosing some utility in one partition element, and $N < \infty$ with no such loss. The comparison of the two cases gives a condition.

PROPOSITION 3. Given $y^* \in (\theta_{n-1}, m_r^*)$ and a partition element p_*, where $m_r^* \in p_*$, the sender keeps $N < \infty$, if the following condition holds:

$$\left(\frac{4\delta}{1+b_p}\right)^2 \leq \left(\frac{|p_*|}{2}\right)^2 + \frac{1-\beta}{1+b_s} - \frac{\sum_{n=1}^{N}(b_s^{n-1})^2}{\left(\sum_{n=1}^{N} b_s^{n-1}\right)^2} \tag{16}$$

The lemma demonstrates the relationship between the parameter of the senders utility b_s and that of the persuader, namely δ and b_p. As far as the optimal low posterior y^* does not have the corner solution at m_r^*, the sender can reduce N to make the condition (16) hold. It has two effects, the first is to increase the third term of the R.H.S., while another effect is a possible increase in size of the partition element with threshold $|p_*|$. The latter is especially significant, if partition is backward and threshold is close to 1. This is a trade-off between the overall decrease in the utility due to less informative partition and removing the payoff loss due to the Bayesian persuasion.

Summing up, the communication occurs in three following stages. In the first stage the sender chooses partition according to the condition (7). Fully truthful cheap talk communication is possible only if $b_s = 1$. Otherwise, fineness of partition decreases with $b_s \to \infty$ or $b_s \to 0$, which gives forward or backward partition, respectively. In the motivating example the sender has a larger utility on smaller values of states, because the taxpayer prefers the tax authority to underestimate the taxable income. The key property of cheap talk is that the sender truthfully reveals the partition element, where $\widehat{\theta} \in p_n$. It reduces the domain of Bayesian persuasion, while cheap talk stays consistent with the Bayesian persuasion taking place in the following stages of the game.

After the cheap talk, the persuader forms a signalling rule. She prefers the trivial signalling rule that reveals the interim posterior with probability 1, if $m_r^* \notin p_n$ or $m_r^* \leq m_r$. In this case Bayesian persuasion makes her strictly worse off. Otherwise, she can increase her payoff, if the condition (12) holds. Because feasible posterior means that can be induced by some signalling rule are restricted, the whole signalling rule is driven by optimal low posterior mean defined in (11). If mentioned condition holds, the persuader applies a non-trivial signalling rule that induces two posterior means. The low posterior mean keeps $a = 0$, as the high posterior mean triggers the action $a = \delta$. The larger the shift δ or the lower the slope b_p, the more a reduced state space is beneficial for the persuader.

5 Conclusion

This paper develops the sequential communication game between three players that integrates cheap talk and Bayesian persuasion. The sender and the persuader communicate using a cheap-talk protocol, while the persuader employs Bayesian persuasion to modify the outcomes of the cheap talk. Considering the motivating example on tax compliance, the tax payer, that is, a cheap-talk sender, reveals her income with some noise as an element of a partition over some interval. This interval can be considered as a range of taxable incomes reported by all taxpayers of similar economic features, so that the tax authority knows the usual magnitude of the income. Since the receiver knows about propensity for under-reporting of income, the tax authority views very low values sceptically. It leads to a larger size of partition elements closer to the lower boundary of possible income.

The foreign tax authority, by contrast, can exploit the tax planning activity. The same as the taxpayer, it benefits from lower values of taxable income, because it attracts taxpayers on its territory and creates demand for services. Hence, the foreign tax authority initiates Bayesian persuasion to convince their colleagues abroad that taxable income is at the lower boundary. In order to do it compellingly, its communication protocol has to be Bayes plausible. When the reported element has low values, intervention of foreign tax authority is nor required. However, when the income passes a specific threshold, Bayesian persuasion becomes beneficial. Moreover, the larger taxes are to be paid into the domestic budget, the more the benefits from the foreign intervention into the communication process.

Bayesian persuasion introduces additional noise into the communication that benefits the foreign tax authority, but reduces the ex-ante payoff of the taxpayer. Despite the interference, the cheap-talk communication remains beneficial for the taxpayer. The more she tends to tax avoidance and higher thresholds for audits, the more the taxpayer condones the interference of the foreign tax authority. In fact, the latter can provide tax preferences that suffices to compensate drop in tax payer's payoff. As a result, the reason for favourable taxation in intermediate countries is not just attraction of investments, but compensation for adverse consequences of additional noise in a communication channel.

The model maybe extended to quadratic functions, which constitute the most widespread form of an agent's utility in cheap-talk models. Another interesting research question is the effect that mediators, who play mixed strategies on cheap-talk stage, have on the following Bayesian persuasion. In the prominent papers on mixed strategies in cheap talk, namely Ivanov (2010) and Ambrus et al. (2013), mediators play mixed strategies over partition elements. By contrast, in our model, the mediator plays mixed strategies within the reported partition element. This difference in approaches needs further research. In addition, a fully algebraic approach of solving Bayesian persuasion problem given in Ivanov (2020) could be also useful.

Appendix

A.1 Proof of Lemma 1

Let's consider reported partition elements $p_n = (\theta_{n-1}, \theta_n]$ and another partition element $p_{n-1} = (\theta_{n-2}, \theta_{n-1}]$ that is adjacent from the left. State θ_{n-1} is the infimum of p_n and the boundary between these intervals. The interim posteriors in the both intervals are $m_r^n = (\theta_{n-1} + \theta_n)/2$ and $m_r^{n-1} = (\theta_{n-2} + \theta_{n-1})/2$. The Bayesian persuasion induces two posterior means $\underline{m}_r$ and $\overline{m}_r$ forming two utilities $u_s(\theta, \underline{m}_r)$ and $u_s(\theta, \overline{m}_r)$. Since the utility is a piece-wise liner function, this equality $\phi u_s(\theta, \underline{m}_r) + (1-\phi) u_s(\theta, \overline{m}_r) = u_s(\theta, m_r^n)$ holds for intervals $(\theta_{n-1}, \underline{m}_r]$ and $[\overline{m}_r, \theta_n]$. For the same reason, on the interval $(\underline{m}_r, \overline{m}_r)$ the convex combination of the utilities picked at the posterior means is a line between values $u_s(\underline{m}_r, m_r^n)$ and $u_s(\overline{m}_r, m_r^n)$. Because the sender's utility is concave, this convex combination makes her strictly worse off.

The worse scenario is, when $\underline{m}_r = \theta_{n-1}$ and $\overline{m}_r = \theta_n$. Meanwhile, the incentive compatible constraint (6) makes the following true $u_s(\theta_{n-1}, m_r^n) = u_s(\theta_{n-1}, m_r^{n-1})$. The same true for another adjacent interval

$p_{n+1} = (\theta_n, \theta_{n+1}]$. As a result, in this case the utility after the Bayesian persuasion is a convex combination of utilities at the boundaries with the adjacent intervals. Meanwhile, due to linear feature of the sender's utility in the interior of p_n the utility $u_s(\theta, m_r^n)$ strictly dominates $u_s(\theta, m_r^{n-1})$ and $u_s(\theta, m_r^{n+1})$. Consequently, the convex combinations is also dominates these utilities for $\theta \in p_n$. Since the Bayesian persuasion keeps the partition incentive compatible in the worse scenario, it is also true for any other Bayes plausible signalling rule chosen by the persuader.

■

A.2 Proof of Lemma 2

When partition element p_n is revealed in the course of cheap talk, the probability of states that do not belong to the element is 0, while the probability of any state within the element is the uniform with density $1/|p_n|$. The receiver's interim posterior m_r^n is always the midpoint of the element, because the uniform distribution of the nature is common knowledge. The geometric structure of Gentzkow and Kamenica (2016) depicted in Figure 3 is an area of feasible distributions confined between two boundaries. The upper boundary is the second integral of the uniform PDF, that is, $\theta^2/(2|p_n|)$. The lower boundary is the second integral of the degenerate density with $\Pr[\theta = m_r^n] = 1$, that is a piece-wise linear function: 0, if $\theta \in (\theta_{n-1}, m_r^n]$, and $-|p_n|/2 + \theta$ otherwise.

As a result, the structures built over any partition element are similar and proportional to size of the element $|p_n|$. The optimal signalling rule with two posteriors is a piece-wise linear function: constant 0, if $\theta \in (\theta_{n-1}, y]$; $f(\theta)$, if $\theta \in (y, \overline{m}_r]$; and $-|p_n|/2 + \theta$ otherwise. Hence, in order to fully define the signalling rule it suffices to find the function $f(\theta)$. To make calculations simpler we normalise the size of p_n to 1, so the normalised low posterior mean is $\tilde{y} = \big(y - \theta_{n-1}\big)/|p_n|$. Next, by construction, the maximum slope of $f(\theta)$ is limited by the upper boundary, so the slope is the derivative $\theta^2/2$. As a result, the required function is:

$$f(\theta) = -\frac{1}{2}\phi(\tilde{y})^2 + \phi(\tilde{y})\theta$$

The next step is to find normalised upper posterior mean $\widetilde{m}_r$. We just need to derive the point of the intersection of $f(\theta)$ and the lower boundary, which is equal to $(1 + \phi(\tilde{y}))/2$. Since we know the normalised low posterior mean $\tilde{y}$ and the normalised upper posterior mean $\widetilde{m}_r$, the Bayes plausibility constraint (5), that is, $\phi(\tilde{y})\tilde{y} + (1 - \phi(\tilde{y})\widetilde{m}_r = \widetilde{m}_r^n$, can be used to derive marginal probability over them. The final step is scaling down of the results using the identity $y = \tilde{y}|p_n| + \theta_{n-1}$.

■

A.3 Proof of Proposition 1

In Bayesian persuasion the persuader receives the expected utility $\nu_p(\theta)$ that is based on Bayes plausible marginal distribution ϕ. She does not know the realised state, so she should maximise her expected value of $\nu_p(\theta)$ based on the uniform distribution over reported element p_n, that is $\mathbb{E}\big[\nu_p(\theta) \mid \theta \sim uni(p_n)\big]$. Since the utility function is piece-wise linear, we can present the expected utility in the following way:

$$\begin{aligned}\mathbb{E}\big[\nu_p(\theta) \mid \theta \sim uni(p_n)\big] &= \phi(\underline{\mu}_r)\mathbb{E}\big[u_p\big(\theta, \underline{m}_r, a = 0\big) \mid \theta \sim uni(p_n)\big] \\ &\quad + \big(1 - \phi(\underline{\mu}_r)\big)\mathbb{E}\big[u_p\big(\theta, \overline{m}_r, a = \delta\big) \mid \theta \sim uni(p_n)\big]\end{aligned} \tag{A.17}$$

For purpose of deriving of an expected value of the persuader's utility function with arbitrary extremum, we define $\underline{m}_r$, $\overline{m}_r$ or m_r^n as x. The expected utility with uniform PDF over reported partition element p_n can be separately defined for each subsets $(\theta_{n-1}, y_n]$ and $(y_n, \theta_n]$.

$$\mathbb{E}\big[u_p(\theta, x, a) \mid \theta \sim uni(p_n)\big] = -\frac{1}{|p_n|}\int_x^{\theta_n}\big[a + b_p(t-x)\big]dt - \frac{1}{|p_n|}\int_{\theta_{n-1}}^{x}\big[a + (x-t)\big]dt \tag{A.18}$$

$$= \frac{1}{|p_n|}\left(x\theta_{n-1} + b_p x\theta_n - \frac{\theta_{n-1}^2}{2} - \frac{b_p\theta_n^2}{2} - (1+b_p)\frac{x^2}{2} - a|p_n|\right)$$

In Lemma 2 both $\overline{m}_r$ and $\phi(\underline{\mu}_r)$ are deterministically defined by $\underline{m}_r$. Hence, the whole expected utility $\mathbb{E}\big[\nu_p(\theta) \mid \theta \sim uni(p_n)\big]$ can be defined as a function of single variable. Then after execution of tedious, but very straightforward calculation, we are able to derive the optimal y^* provided in the proposition. Due to feasibility reasons, the optimal high posterior mean has to be in range $(\theta_{n-1}, m^*]$ for given partition element p_n. It cannot be less then the low boundary of the element, because $\Pr[\theta \notin p_n] = 0$. Meanwhile, it cannot be larger then threshold m_r^*, because then the utilities in both posteriors will be shifted by δ.

■

A.4 Proof of Proposition 2

Regarding the condition that makes Bayesian persuasion beneficial. The first, we need to make substitutions in the inequality (9) using $\overline{m}_r$ and $\phi(\underline{\mu}_r)$ from Lemma 1 and the expressions (A.17) and (A.18). The second, we should do another straightforward algebra, that provides the condition.

■

A.5 Proof of Proposition 3

The inference is based on comparison of two values. The first is ex-ante utility the sender receives, if she applies $N = \infty$ and the persuader applies non-trivial signalling rule. The ex-ante utility for this case, before a signalling rule is applied, is (15). Hence, we just need to evaluate loss of utility in the partition element that includes threshold m_r^*. It is difference between ex-ante utility for the case of $u_s(\theta, m_r^n)$ and the case of ν_s, that is similar to ν_p, but both utilities at posteriors have $a = 0$. Using optimal y^* for the case $y^* < m_r^*$ a bit of algebra gives the following ex-ante utility:

$$-\frac{1-\beta}{8} - (1+b_s)\left(\frac{|p_*|^2}{32} - \frac{2\delta^2}{(1+b_p)^2}\right)$$

The second component for the comparison is ex-ante utility (14). Rearranging and cancelling out of a few of factors gives the condition (16).

■

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